MQIL Dataset Project

Feature Extraction – Phase 2

Purpose: Explain how feature_extractor.cpp reads generated C programs, extracts numerical features, and stores them in features.csv.

1. Overall Workflow

```
Generated C Programs → Read Source Code → Extract Features → Determine Label → Write features.csv
```

The extractor processes every .c file in the dataset directory. Each program becomes one row in features.csv.

2. Feature Categories

Category	Features
Basic	lines, has_secret, has_public
Control flow	if_count, else_count, for_count, while_count, switch_count, case_count
Operations	return_count, assignment_count, comparison_count, arithmetic_count, array_access_count, function_count
Information flow	secret_value, public_value, direct_secret_use, indirect_secret_use, loop_secret_use, branch_secret_use
Derived	complexity

3. Basic Features

lines: Counts the number of newline characters in the source program.
has_secret: Checks whether an integer variable named secret exists; returns 1 or 0.
has_public: Checks whether an integer variable named public exists; returns 1 or 0.

4. Control-Flow Features

The countKeyword() function counts occurrences of C control keywords using regular expressions.

Feature	Meaning
if_count	Number of if statements
else_count	Number of else statements
for_count	Number of for loops
while_count	Number of while loops
switch_count	Number of switch statements
case_count	Number of case statements

5. Operation and Language Features

Feature	Meaning
return_count	Counts return
assignment_count	Counts assignment = patterns
comparison_count	Counts ==, !=, <=, >=, < and >
arithmetic_count	Counts +, -, * and /
array_access_count	Detects patterns such as arr[0] or arr[i]
function_count	Detects function-definition patterns

6. Secret and Public Values

extractVariable() searches for integer declarations and extracts their numeric values.

```
int secret = 53; → secret_value = 53
int public = 51; → public_value = 51
```

7. Information-Flow Features

direct_secret_use: Counts occurrences of the keyword secret in the current implementation.
indirect_secret_use: Detects a variable initialized from secret, e.g., int temp = secret;
loop_secret_use: Detects secret in a for/while condition.
branch_secret_use: Detects secret in an if/else-if condition.

8. Complexity

```
complexity = 1 + if_count + for_count + while_count + switch_count
```

9. Label Assignment

Labels are determined from filenames. The implemented categories include no_leak, direct_leak, branch_leak, loop_leak, indirect_leak, and complex. If no pattern matches, the default label is unknown.

10. Output

After extraction, all values are written in a fixed order to features.csv. Each generated C program produces one row.

11. Presentation Summary

• The extractor automatically processes every generated C program.
• It uses keyword matching and regular expressions.
• It extracts structural, control-flow, operation-based, and information-flow features.
• Every program is converted into a numerical feature vector.
• The feature vectors are stored in features.csv for the next phase.

Conclusion

Phase 2 transforms raw C programs into structured numerical data. This provides a consistent representation for further analysis and future machine-learning stages of the MQIL project.